

BTIT 304

III Semester Examination, Nov. - Dec., 2017

B.Tech. / B.Tech. + M.Tech. / B.Tech. + MBA [IT]

Fundamentals of Operating System

Choice Based Credit System (CBCS)

Time: 3 Hrs.

Maximum Marks : 60

Minimum Pass Marks: 24

- Note: (1) All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks and part 'C' carries 6 marks.
(2) From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
(3) Draw the neat diagram, wherever necessary.
(4) Assume suitable data, wherever necessary.

- Q.1.(A)** Define Operating System? Explain with diagram. 03
(B) What are evaluations of operating system currently? Explain it. 03
(C) List out most widely used types of Operating system. Explain them. 06

OR

What are the different types of function performed by an OS? Explain each.

- Q.2.(A)** What is process? Explain process state diagram. 03
(B) What is system call? Define it with example. 03
(C) Explain operating system structure in detail, according to simple, layered and virtual machine. 06

OR

Explain storage hierarchy and different type of protection in operating system.

- Q.3.(A)** Define thread. Differentiate between user level and kernel level thread. 03
(B) Discuss Banker's Algorithm. Explain deadlock prevention technique. 03
(C) What is semaphore? How can it solve the problem of critical section? 06

OR

Contd.....

Consider the following Job

Job	Burst Time
1	10
2	29
3	3
4	7
5	12

Consider the FCFS, SJF and RR (quantum =10) scheduling algorithm for this set of Process. Which algorithm would give the minimum average waiting time?

Q.4.(A) Differentiate between physical and logical address space. 01

(B) What is contiguous memory allocation? Describe different allocation strategy. 02

(C) Consider the following page reference strings :

1,2,3,4,5,3,4,1,6,7,8,7,8,9,7,8,9

How many page faults would occur for the FIFO page replacement Algorithm, assuming 2, 4 and 5 frames being made available? 06

OR

The disk queue in the request for I/O to block, on cylinders are 98, 183, 37, 122, 14, 124, 65, 67. If the disk head is initially at 53 compute the total head movement for the following algorithm.

(a) FCFS (b) SCAN

Q.5.(A) What are the goals of protection in operating system? 02

(B) Explain domain protection. 03

(C) How can we achieve authentication in operating system for security? 06

OR

What is one time password method? How can we implement it for OS security?

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